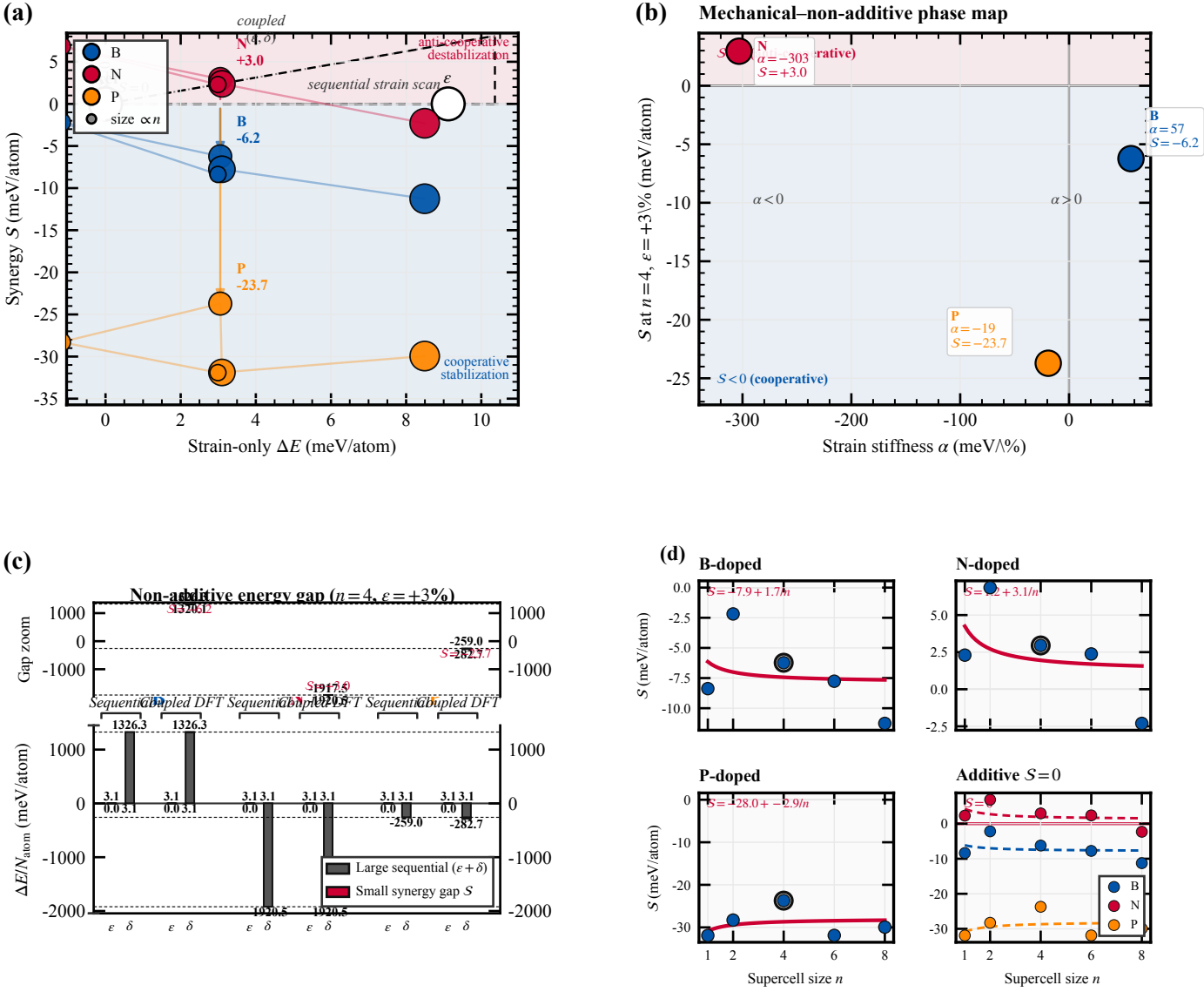

(a)
coupled
(ε, δ)
anti-cooperative
destabilization
sequential strain scan ε
B
N
P
size ∝ n
N
+3.0
B
-6.2
P
-23.7
cooperative
stabilization
Synergy S (meV/atom)
Strain-only ΔE (meV/atom)
(b)
Mechanical–non-additive phase map
N
α = −303
S = +3.0
B
α = 57
S = −6.2
α < 0
α > 0
P
α = −19
S = −23.7
S < 0 (cooperative)
S at n = 4, ε = +3% (meV/atom)
Strain stiffness α (meV/%)
(c)
Non-additive energy gap (n = 4, ε = +3%)
Gap zoom
Sequential
Coupled DFT
Large sequential (ε + δ)
Small synergy gap S
ΔE/N_atom (meV/atom)
(d)
B-doped
S = −7.9 + 1.7n
N-doped
S = 2 + 3.1/n
P-doped
S = −28.0 + −2.9/n
Additive S = 0
S (meV/atom)
Supercell size n